

Macro con Agentes Heterogéneos

Lian Allub

03/09/2021

Decisiones Ocupacionales

- Quadrini (RED, 2000): Entrepreneurship, Savings and Social Mobility
- Allub y Erosa (JME, 2019): Financial Frictions, Occupational Choice and Economic Inequality

Quadrini (RED, 2000):
Entrepreneurship, Savings and Social Mobility

Quadrini: Introducción

- La distribución de riqueza en US es más concentrada que la distribución de ingresos
- Modelos de riesgo laboral idiosincrático no son capaces de generar la desigualdad de riqueza observada en los datos
 - ▶ Especialmente no pueden generar la concentración en el tope de la distribución de riqueza

Quadrini: Introducción

- Estudia el rol que juega el emprendedurismo para explicar la concentración de riqueza
 - ▶ ¿Es el emprendedurismo relevante para caracterizar las decisiones de acumulación diferentes de los agentes localizados en el tope de la distribución del ingreso?
 - ▶ Si es diferente, ¿es cuantitativamente relevante para generar la concentración observada en los datos?
- Estudia también el rol del emprendedurismo para explicar patrones de movilidad social
 - ▶ En los datos los emprendedores presentan mayor movilidad ascendente
 - ▶ Debido a restricciones de crédito o costos de financiamiento elevado, el emprendedurismo es menos probable para agentes en la cola inferior de la distribución de riqueza y por lo tanto tienen menos probabilidad de ascender en la distribución

Quadrini: Introducción

- El modelo tiene 3 razones para ahorrar para empresarios
 - 1 Acumular el capital mínimo requerido para llevar a cabo la actividad empresarial
 - 2 Mayor riesgo financiero que trabajadores
 - 3 Costo de pedir prestado. La tasa de préstamos es mayor a la de depósitos

Quadrini: Datos

- Empresarios: Familias que son dueñas o tienen interés financiero en un emprendimiento.
- Empresarios II: Familias donde el jefe de hogar es un self-employed en su principal trabajo.
- Los resultados son similares, por lo que van a trabajar con la primer definición.

Quadrini: Emprendedurismo y concentración de riqueza

TABLE I
Distribution of U.S. Household Wealth and Income

	Top percentiles					Gini index	Negative and zero
	1%	5%	10%	20%	30%		
Wealth							
PSID 1984	30.0	49.2	61.7	76.6	85.8	0.76	10.6
PSID 1989	25.4	47.0	60.9	77.1	86.9	0.76	12.3
PSID 1994	22.6	44.8	59.1	75.9	85.9	0.75	12.9
SCF 1989	35.7	58.0	70.1	83.7	91.8	0.86	11.7
SCF 1992	29.5	53.5	66.1	79.5	87.6	0.78	6.9
Income							
PSID 1984	7.5	19.4	30.2	46.9	60.0	0.43	0.5
PSID 1989	8.1	20.6	31.6	48.2	61.0	0.45	0.5
PSID 1992	7.2	19.9	31.1	48.4	61.7	0.45	0.7
SCF 1988	16.9	31.7	42.3	57.2	68.8	0.54	0.7
SCF 1991	18.5	34.4	45.1	59.9	70.9	0.57	1.2

- Gran concentración de riqueza e ingresos en los percentiles más altos
- Mayor concentración en riqueza que en ingresos
- Notar la mayor concentración en la SCF. Esto es porque la PSID no capta bien el top income.

Quadrini: Emprendedurismo y concentración de riqueza

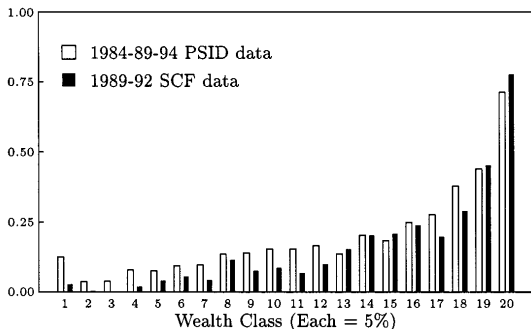


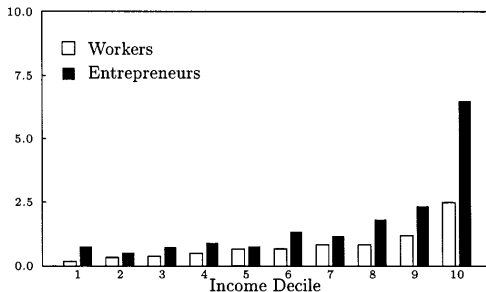
FIG. 1. Percentage of business families over wealth classes: Each class includes 5% of all families.

- La cantidad de empresarios aumenta en los percentiles más altos de la distribución de riqueza

Quadrini: Emprendedurismo y concentración de riqueza

- El gráfico muestra la tenencia de riqueza de trabajadores y empresarios para diferentes deciles de ingresos como proporción de la riqueza promedio

A - Average 1984-89-94 PSID data



- Los empresarios tienen en promedio mayores niveles de riqueza en relación a su ingreso que los trabajadores
- Empresarios 14% del total. Poseen 22% del ingreso pero 40% de la riqueza
- Hay una concentración de riqueza que no está explicada por concentración de ingresos

Quadrini: Emprendedurismo y movilidad social

TABLE II
Five-Year Transition Matrices for Net Family Wealth

(A) 1984–1989 transition

	Class I	Class II	Class III	Class I	Class II	Class III
	Staying workers			Switching workers		
Class I	0.81	0.17	0.02	0.52	0.31	0.17
Class II	0.22	0.65	0.13	0.12	0.51	0.37
Class III	0.02	0.22	0.76	0.00	0.20	0.80
	Switching entrepreneurs			Staying entrepreneurs		
Class I	0.81	0.14	0.05	0.25	0.49	0.26
Class II	0.23	0.58	0.19	0.17	0.37	0.46
Class III	0.01	0.21	0.78	0.02	0.09	0.89

- En la clase más baja, la probabilidad de ascender de clase es mayor para los trabajadores (W) que se cambian a empresarios (E) que para los que permanecen como trabajadores
- En la clase del medio, para los W que cambian E la probabilidad de ascender de clase es mayor que la de bajar de clase. Lo opuesto ocurre para los W que permanecen
- Clase alta, la probabilidad de caer de clase es mayor para los que permanecen como W que para los que pasan de W a E.

Quadrini: Emprendedurismo y movilidad social

- Para las familias que eran trabajadores en t
 - ▶ En la clase más baja, la probabilidad de ascender de clase es mayor para los trabajadores (W) que se cambian a empresarios (E) que para los que permanecen como trabajadores
 - ▶ En la clase del medio, para los W que cambian E la probabilidad de ascender de clase es mayor que la de bajar de clase. Lo opuesto ocurre para los W que permanecen
 - ▶ Clase más alta, la probabilidad de caer de clase es mayor para los que permanecen como W que para los que pasan de W a E.
- Para las familias que eran empresarias en t
 - ▶ En la clase más baja, la probabilidad de ascender de clase es mayor para los E que permanecen como E que los que cambian a W
 - ▶ En la clase del medio, la probabilidad de ascender de clase es mayor que la de bajar de clase para los E que permanecen como E. Lo opuesto ocurre para los que pasan de E a W
 - ▶ Clase más alta, la probabilidad de caer de clase es mayor para los que cambian a W que para los que permanecen E.

Quadrini: Persistencia y rotación de empresarios

- Hipótesis → Las conductas de ahorro de las familias cambian al volverse empresarias
- Este cambio hace que acumulan más riqueza y asciendan de clase más rápido
- Rol de la persistencia → mientras más persistente sea la actividad empresarial mayor será la acumulación de riqueza

Quadrini: Persistencia y transición de empresarios

TABLE IV
Exit Rates from Entrepreneurship (Top Section) and Entrance Rates to Entrepreneurship
(Bottom Section)

	Exit rate	No. of families ^a
(a) Business owners		
All business families	24.2	522
With one year of entrepreneurial tenure	44.7	151
With two years of entrepreneurial tenure	30.8	80
With three or more years of entr. tenure	13.4	291
(b) Self-employed		
All business families	13.6	384
With one year of entrepreneurial tenure	35.2	75
With two years of entrepreneurial tenure	19.1	48
With three or more years of entr. tenure	7.2	261
	Entrance rate	No. of families ^a
(a) Business owners		
All worker families	3.7	4722
Without entrepreneurial experience	2.6	4506
With entrepreneurial experience	23.1	216
(b) Self-employed		
All worker families	2.9	2837
Without entrepreneurial experience	2.0	2556
With entrepreneurial experience	27.2	281

Note. Annual values averaged over the sample period 1973–1992.

^aThe number of families is the average sample size in each year, from 1973 through 1992.

- Tasa de salida es alta para empresarios nuevos, pero cae rápido
 - ▶ Aprendizaje en la actividad empresarial
- Las tasas de entrada al emprendedurismo son más altas para familias con experiencia en actividades empresarias
- Suma de estos 2 flujos → Rotación de empresarios baja y persistencia alta

Quadrini: Modelo

- Continuo de agentes que viven infinitos períodos (masa 1)
- Cada período eligen si ser empresarios o proveer trabajo en el mercado
- 3 sectores: Familias, Sector Productivo, Sector de Intermediación
- Familias:
 - ▶ Maximizan la utilidad esperada: $E_0\{\sum_{t=0}^{\infty} \beta^t u(c_t)\}$
 - ▶ β es el factor de descuento.
- Habilidad de trabajo
 - ▶ Tiene una dotación de unidades de eficiencia $\varepsilon \in \mathcal{E} = \{\varepsilon_1, \dots, \varepsilon_{N_\varepsilon}\}$
 - ▶ Estas unidades se pueden usar en su propia empresa u ofrecerlas en el mercado por un salario w
 - ▶ Trabajo es igual de productivo en su propio negocio o en otro negocio
→ no importa donde trabaje → ofrecen todas las unidades al mercado
 - ▶ La variable ε se conoce al final del período y sigue un proceso de Markov de primer orden con transición $\Gamma(\varepsilon', \varepsilon)$

Quadrini: Modelo

- Oportunidad Empresarial:

- ▶ Además de proveer trabajo en el mercado, los agentes pueden ser empresarios
- ▶ Los agentes reciben ideas de negocios κ al final de cada período
- ▶ κ se saca de $\mathcal{K} = \{0, k_1, k_2, \dots, k_{N_k}\}$
- ▶ El primer elemento corresponde a no recibir ideas
- ▶ La nueva idea empresarial junto con el proyecto implementado el período actual, forman el conjunto de proyectos con el cual la familia puede empezar un negocio el período siguiente.
- ▶ La variable κ es estocástica y se distribuye de acuerdo $P_k(\kappa)$, donde k denota el proyecto implementado el período actual
- ▶ La dependencia de P de k formaliza la hipótesis que hay un proceso de aprendizaje en la actividad empresarial

Quadrini: Modelo

- Sector Productivo:

- ▶ 2 sectores:
 - ★ No corporativo: unidades pequeñas (empresarios)
 - ★ Corporativo: unidades grandes
- ▶ Empresarios tienen 2 características:
 - ★ Tienen riesgo empresarial idiosincrático → la riqueza de la familia depende del resultado del negocio
 - ★ Enfrentan restricciones al crédito → Dotación de capital de las firmas dependen de la riqueza
- ▶ Sector corporativo:
 - ★ Función de producción de CRS
 - ★ $Y_c = K_c^\theta N_c^{1-\theta}$
 - ★ Y_c es producción, K_c capital, N_c unidades de trabajo eficientes
 - ★ El capital se deprecia a una tasa δ

Quadrini: Modelo

- Empresarios:

- ▶ Cada período las familias reciben una idea κ
- ▶ El capital requerido para implementar esta idea es indivisible
- ▶ Función de producción: $y = \eta^\nu k^\nu n^{1-\nu}$ con $0 \leq \nu \leq 1$
 - ★ y es producción bruta (producción+ capital no depreciado)
 - ★ n son unidades de trabajo de eficientes
 - ★ η shock tecnológico. Proceso de markov con transición $Q(\eta'|\eta)$
- ▶ El conjunto del cual proviene η así como su distribución de probabilidades dependen del proyecto implementado k
- ▶ El primer elemento η_1 es un shock malo con alta persistencia
- ▶ Si toca η_1 el empresario abandona el negocio
- ▶ Secuencia:
 - ① Al final del período se elige el proyecto k a implementar del conjunto disponible
 - ② Al inicio del próximo período se observa η y se decide n
- ▶ Un empresario siempre puede continuar con el proyecto actual.
 - ★ El conjunto de ideas disponibles se compone del proyecto actual y de la nueva idea

Quadrini: Modelo

- Intermediarios Financieros:

- ▶ Reciben depósitos (pagan r_D) y dan préstamos $r_L = r_D + \phi$
- ▶ La actividad de préstamos es una de CRS con un costo proporcional por unidad intermediada
- ▶ Para el sector corporativo este costo es 0, mientras que para empresarios es $\phi > 0$
- ▶ El intermediario puede determinar un monto máximo tal que siempre convenga repagar $\Rightarrow$ No hay default
- ▶ Los recursos disponibles al final del período de un proyecto k con η_{min}
 $DR_{min} = \max_n \{ \eta_{min}^\nu k^\nu n^{1-\nu} - nw \} + \varepsilon w$ (se asume $k > a$)
- ▶ Para un empresario, como $k > a$, la tasa relevante es r_L
- ▶ Lo que tendrá que pagar por el préstamo es $(k - a)(1 + r_L)$
- ▶ Lo que tiene que pagar tiene que ser menor a DR_{min} , lo que implica
 $a > k - \frac{DR_{min}}{1+r_L}$

Quadrini: Modelo

- Costo del capital y beneficios:

- Si un agente decide ser empresario e implementar un proyecto k , el siguiente período luego de observar η decide el trabajo n maximizando

$$\pi(a, k, \eta) = \max_n \{ \eta^\nu k^\nu n^{1-\nu} - nw - (1+r)k \}$$

- donde

$$r = \begin{cases} r_D & \text{si } k \leq a \\ r_D + \phi\left(\frac{k-a}{k}\right) & \text{si } k > a \end{cases}$$

- La firma toma r_d, r_L, w como dados $\Rightarrow n(k, \eta) = \eta k \left(\frac{1-\nu}{w}\right)^{1/\nu}$
- Substituyendo obtenemos $\pi(a, k, \eta) = \nu \eta k \left(\frac{1-\nu}{w}\right)^{(1-\nu)/\nu} - (1+r)k$

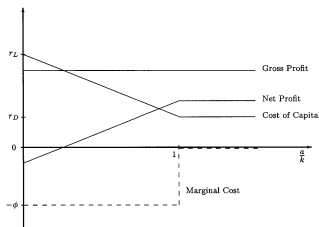


FIGURE 3

Quadrini: Modelo

- Value function al final del período:

$$\tilde{v}(\varepsilon, a, k, \eta, \kappa, \varepsilon') = \max_{a', k' \in \{k, \kappa\}} u(c) + \beta \sum_{\eta'} v(\varepsilon'(a', k', \eta')) Q_k(\eta'/\eta) \quad (9)$$

subject to

$$c = a(1 + r_D) + \pi(a, k, \eta) + \varepsilon w - a'$$

$$a' \geq k' - \frac{\nu \eta_{\min} k' ((1 - \nu)/w)^{(1-\nu)/\nu} + \varepsilon' w}{1 + r_L}.$$

- Value function al inicio del período

$$v(\varepsilon, a, k, \eta) = \sum_{\kappa, \varepsilon'} \tilde{v}(\varepsilon, a, k, \eta, \kappa, \varepsilon') P_k(\kappa) \Gamma(\varepsilon'/\varepsilon). \quad (10)$$

Quadrini: Calibración

- Preferencias:

- ▶ Función de utilidad: $u(c_t) = c_t^{1-\sigma} / (1 - \sigma)$. Se elige $\sigma = 2$
- ▶ β se calibra para que en equilibrio $r_D = 3.5\%$
 - ★ Promedio entre la tasa libre de riesgo y el retorno promedio de activos riesgosos

Quadrini: Calibración

- Habilidades como trabajador

- ▶ Sigue un proceso de Markov de primer orden con 4 estados
- ▶ La matriz de transición es Γ
- ▶ Existe una probabilidad $p = 1/35$ de reemplazar la generación actual por una nueva.
- ▶ La habilidad de cada generación sigue un proceso de Markov de 2 estados con transición Γ_ε
- ▶ Cada generación puede ser 2 tipos:
 - ★ Tipo 1 toma valores $\{\varepsilon_{11}, \varepsilon_{12}\}$
 - ★ Tipo 2 toma valores $\{\varepsilon_{21}, \varepsilon_{22}\}$
- ▶ La probabilidad de que tu descendiente conserve tu tipo es de 0.75 $\rightarrow$ Correlación intergeneracional de ingresos 0.5
- ▶ La matriz de transición entre tipo de ingresos (considerando p y la probabilidad de conservar la habilidad)

$$\Pi = \begin{Bmatrix} 0.9929 & 0.0071 \\ 0.0071 & 0.9929 \end{Bmatrix}.$$

- ▶ $\Gamma = \Pi \otimes \Gamma_\varepsilon$

Quadrini: Calibración

- Habilidades como trabajador (continuación)
 - ▶ Necesita calibrar Γ_ε , $\{\varepsilon_{11}, \varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22}\}$
 - ▶ Asume que $\ln(\varepsilon_{i,t+1}) = \alpha_i + \rho \ln(\varepsilon_{i,t}) + v_{t+1}$ con $v_{t+1} \sim N(0, \sigma_v^2)$
 - ★ i es el tipo de habilidad de la persona
 - ★ α_i es el parámetro que caracteriza a la media del proceso de ingresos para la familia de tipo i
 - ▶ $\rho = 0.496$ y $\sigma_v = 0.332$ se calibran usando datos de la PSID de ingresos.
 - ▶ La habilidad de una familia con un α_i determinado se aproxima con un proceso de Markov con 2 estados y transición $\Gamma_\varepsilon(\varepsilon', \varepsilon)$ simétrica.
 - ▶ Los momentos usados para calibrar Γ_ε son:
 - ★ La media no condicionada de $\ln(\varepsilon)$ igual a $\alpha_i / (1 - \rho)$
 - ★ La autocorrelación ρ
 - ★ La desviación estándar, $\sigma_v / \sqrt{1 - \rho^2}$
 - ★ α_i es el parámetro que caracteriza a la media del proceso de ingresos para la familia de tipo i
 - ▶ α_i se calibra para matchear un Gini de ingresos de 0.38

Quadrini: Calibración

- Tecnología de producción

- ▶ Proporción de capital del sector no corporativo $\rightarrow 0.40$
- ▶ El capital de ambos sectores se deprecia a una tasa $\delta = 0.062$
- ▶ Ley de movimiento del capital $K_{t+1} = (1 - \delta)K_t + I_t$
- ▶ El sector corporativo produce $Y = K^\theta L^{1-\theta}$ con $\theta = 0.33$
- ▶ Sector no corporativo:
 - ★ Existen 3 proyectos k_1, k_2, k_3
 - ★ Para calibrarlos se usa la distribución de riqueza de los empresarios.
 - ★ Asigna probabilidades 0.6, 0.3, 0.1 respectivamente para capturar la asimetría de la distribución
 - ★ Con esto, fijamos $k_2/k_1 = 10$ y $k_3/k_1 = 100$
 - ★ η toma 2 valores $\{\eta_1, \eta_2\}$ y tiene matriz de transición $Q_k(\eta'|\eta)$
 - ★ El primer componente es muy persistente $Q_k(\eta'_1|\eta_1) = 1$ y lo suficientemente bajo.
 - ★ Solo queda determinar 1 valor para cada Q_k para $k \in \{k_1, k_2, k_3\}$
 - ★ $Q_1(\eta'_1|\eta_2) = 0.25$, $Q_2(\eta'_1|\eta_2) = 0.08$ y $Q_3(\eta'_1|\eta_2) = 0.03$. Esto da una tasa de salida promedio de 18%
- ▶ w y r provienen de la CPO del sector corporativo

Quadrini: Calibración

- Tecnología de producción
 - ▶ Sector no corporativo:
 - ★ Para calibrar $P_k(\kappa)$ asume que solo puede arribar una idea inmediatamente superior
 - ★ Como nunca eligirá reducir de tamaño, la probabilidad del último k es irrelevante
 - ★ $\bar{\eta}$ y ν se eligen para matchear la participación de empresarios en el ingreso
 - ★ η_1 y η_2 de cada proyecto se eligen para matchear la volatilidad del ingreso empresarial
 - ★ Tener en cuenta que los empresarios más experimentados tienen negocios menos riesgosos
 - ▶ w y r provienen de la CPO del sector corporativo
- Tecnología de intermediación
 - ▶ Se usa el spread para disciplinar $r_L - r_D = \phi = 0.045$

TABLE VII
Calibration Values for the Baseline Model

Intertemporal discount rate	β	0.934
Relative risk aversion parameter	σ	2.000
Intermediation cost	ϕ	0.045
Corporate capital income share	θ	0.330
Depreciation rate	δ	0.062
Noncorporate size projects	k	$\{ 1.63 \quad 16.26 \quad 162.61 \}$
Mean value of the shock	$\bar{\eta}$	2.374
Values of the shock	η	$\left\{ \begin{array}{l} 1.888 \quad 2.536 \\ 1.740 \quad 2.429 \\ 1.842 \quad 2.390 \end{array} \right\}$
Probability distribution of the shock	$Q_k(\eta'/\eta_2)$	$\left\{ \begin{array}{l} 0.250 \quad 0.750 \\ 0.080 \quad 0.920 \\ 0.030 \quad 0.970 \end{array} \right\}$
Arrival probability of a new project	$P_k(\kappa)$	$\left\{ \begin{array}{l} 0.024 \\ 0.110 \\ 0.075 \end{array} \right\}$
Values of labor ability	ε	$\{ 0.334 \quad 0.720 \quad 1.390 \quad 2.989 \}$
Transition probabilities for the labor ability	$\Gamma = \Pi \otimes \Gamma_\varepsilon$	$\left\{ \begin{array}{l} 0.743 \quad 0.250 \quad 0.005 \quad 0.002 \\ 0.250 \quad 0.743 \quad 0.002 \quad 0.005 \\ 0.005 \quad 0.002 \quad 0.743 \quad 0.250 \\ 0.002 \quad 0.005 \quad 0.250 \quad 0.743 \end{array} \right\}$

Quadrini: Resultados

- Se usa el modelo para analizar cuantitativamente cuán importante es el emprendedurismo para explicar la concentración de la riqueza y la movilidad social
- 2 conjunto de resultados
 - 1 Se evalúa la capacidad del modelo para generar la desigualdad de riqueza y movilidad social entre empresarios y trabajadores observada en los datos
 - 2 Se evalúa la importancia de los empresarios para generar estos patrones.
 - ★ Se compara el benchmark con un modelo sin empresarios
 - 3 Ejercicios de sensibilidad para ciertos parámetros

Quadrini: Resultados: Tenencia de activos y movilidad de empresarios y trabajadores

TABLE VIII
Wealth-to-Income Ratios for Workers and Entrepreneurs

	Workers		Entrepreneurs	
	% of househ.	Wea.-inc. ratio	% of househ.	Wea.-inc. ratio
Model economy				
Income Class I	31.0	1.32	2.3	12.51
Income Class II	30.1	2.41	3.3	2.48
Income Class III	26.9	3.04	6.4	5.36
Total	88.0	2.68	12.0	5.15
PSID data				
Income Class I	31.4	3.74	2.0	11.68
Income Class II	29.7	2.82	3.6	4.52
Income Class III	25.2	2.71	8.1	5.90
Total	86.3	2.86	13.7	5.83

Note. Model economy and average 1984, 1989, and 1994 PSID data.

- Diferencia en los ratios Riqueza-Ingreso entre trabajadores y empresarios más bajos en el modelo

Quadrini: Resultados: Tenencia de activos y movilidad de empresarios y trabajadores

TABLE IX
Distribution of Agents among Wealth Classes

	Model economy		PSID data	
	% of workers	% of entrepr.	% of workers	% of entrepr.
Wealth Class I	31.6	1.7	31.6	1.8
Wealth Class II	29.4	3.9	29.8	3.5
Wealth Class III	27.0	6.4	24.9	8.4
Total	88.0	12.0	86.4	13.6
Neg. & zero	15.5	0.6	11.3	0.6

Note. Model economy and average 1984, 1989, and 1994 PSID data.

- El modelo captura bien la concentración de empresarios en el grupo de riqueza más alta

Quadrini: Resultados: Tenencia de activos y movilidad de empresarios y trabajadores

TABLE X
Transition Matrices for Net Family Wealth, Five-Period Simulation

	Class I	Class II	Class III	Class I	Class II	Class III
	Staying workers			Switching workers		
Class I	0.81	0.19	0.00	0.61	0.38	0.01
Class II	0.22	0.64	0.14	0.13	0.71	0.16
Class III	0.00	0.18	0.82	0.00	0.15	0.85
	Switching entrepreneurs			Staying entrepreneurs		
Class I	0.77	0.23	0.00	0.40	0.58	0.02
Class II	0.23	0.66	0.11	0.03	0.64	0.33
Class III	0.00	0.15	0.85	0.00	0.02	0.98

TABLE II
Five-Year Transition Matrices for Net Family Wealth
(A) 1984–1989 transition

	Class I	Class II	Class III	Class I	Class II	Class III
	Staying workers			Switching workers		
Class I	0.81	0.17	0.02	0.52	0.31	0.17
Class II	0.22	0.65	0.13	0.12	0.51	0.37
Class III	0.02	0.22	0.76	0.00	0.20	0.80
	Switching entrepreneurs			Staying entrepreneurs		
Class I	0.81	0.14	0.05	0.25	0.49	0.26
Class II	0.23	0.58	0.19	0.17	0.37	0.46
Class III	0.01	0.21	0.78	0.02	0.09	0.89

- El modelo captura cualitativamente bien los patrones de movilidad
- Empresarios tienden a permanecer o subir de clase mientras que trabajadores tienden a permanecer o bajar

Quadrini: Resultados: Emprendedurismo y concentración de la riqueza

TABLE XI
Percentage of Total Wealth and Income Held by Percentile Groups and Gini Indexes

	Top percentiles					Gini index	Zero & neg.
	1%	5%	10%	20%	30%		
Wealth							
Model economy	24.9	45.8	57.1	73.2	84.0	0.74	15.9
PSID data	26.0	47.0	60.6	76.5	86.2	0.76	11.9
Only workers	4.2	15.3	26.2	44.5	58.3	0.55	10.1
Income							
Model economy	7.9	18.2	28.5	46.8	64.0	0.45	0.1
PSID data	7.6	19.9	30.9	47.8	60.9	0.44	0.6
Only workers	3.8	13.4	24.4	45.7	60.2	0.42	0.0

- Riqueza

- ▶ Benchmark es capaz de captar la desigualdad observada en los datos
- ▶ Sin empresarios no genera suficiente concentración en los percentiles más altos

- Ingreso

- ▶ Benchmark captura bien la desigualdad observada en los datos
- ▶ Sin empresarios falla en particular en el top 5% y 1%

Quadrini: Resultados: Emprendedurismo y concentración de la riqueza

- Sin empresarios la única razón para ahorrar es la suavización de consumo
 - ▶ Después de cierto nivel de riqueza, los incentivos a ahorrar son bajos
- ¿Qué explica la mayor concentración de riqueza en este modelo?
 - ▶ Aprovechar la oportunidad si toca un buen proyecto empresarial
 - ▶ Costo más alto de pedir prestado
 - ▶ Riesgo empresarial idiosincrático.
 - ★ El ingreso del empresario dependerá del shock que es desconocido al momento de decidir $\Rightarrow$ Lleva a ahorrar más por razones precautorias
 - ▶ Selección
 - ★ Por las restricciones al crédito los empresarios provienen de los trabajadores más ricos
 - ★ Solo los empresarios exitosos subsisten $\Rightarrow$ acumulan más riqueza
 - ▶ Persistencia y rotación de los empresarios

Quadrini: Resultados: Ejercicios de Sensibilidad

- 2 Ejercicios
 - ▶ Sin costos de intermediación
 - ▶ Diferentes niveles de riesgo empresarial
- Sin costo de intermediación:
 - ▶ Menor concentración de riqueza
- Cambios en el riesgo empresarial
 - ▶ Menos riesgo: Disminuye la concentración de la riqueza
 - ▶ Más riesgo: Aumenta la concentración de la riqueza

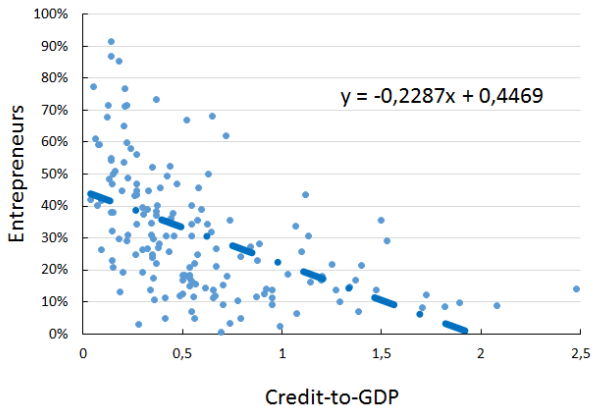
**Allub y Erosa (JME, 2019):
Financial Frictions, Occupational Choice and
Economic Inequality**

Motivation

- What accounts for differences in **productivity** across countries?
- The role of financial frictions in the misallocation of resources
 - ▶ Erosa (2001), Jeong and Townsend (2007), Amaral and Quintin (2010), Buera, Kaboski, and Shin (2010), Buera and Shin (2011), Greenwood, Sanchez, and Wang (2007), Clementi, McDonald and Rui Castro, Midrigan and Xu (2014) and many more...
- However, these theories are inconsistent with key facts on entrepreneurship in developing countries

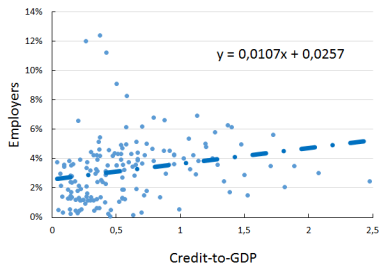
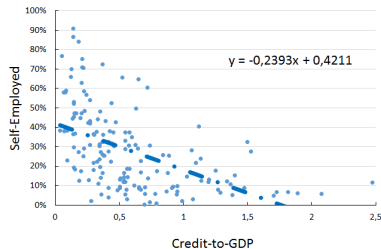
Occupational Structure

- Negative relation between entrepreneurship and credit...



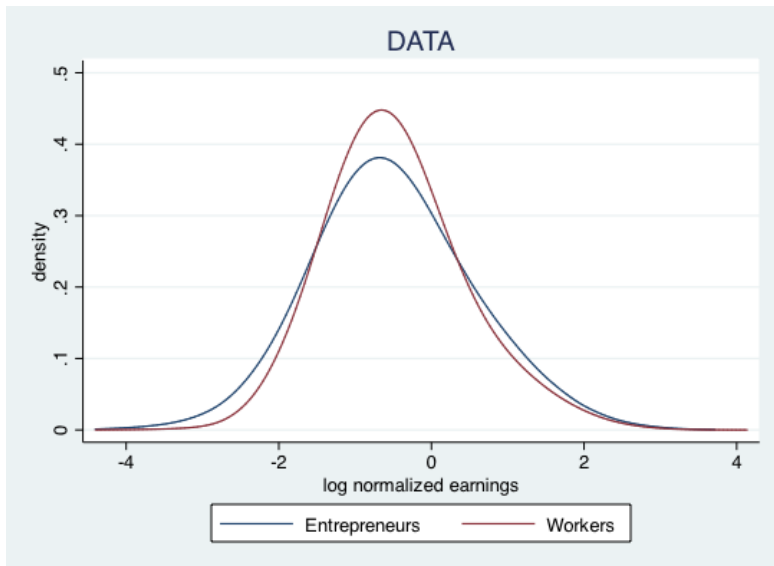
Occupational Structure

- ...mostly explained by self-employment



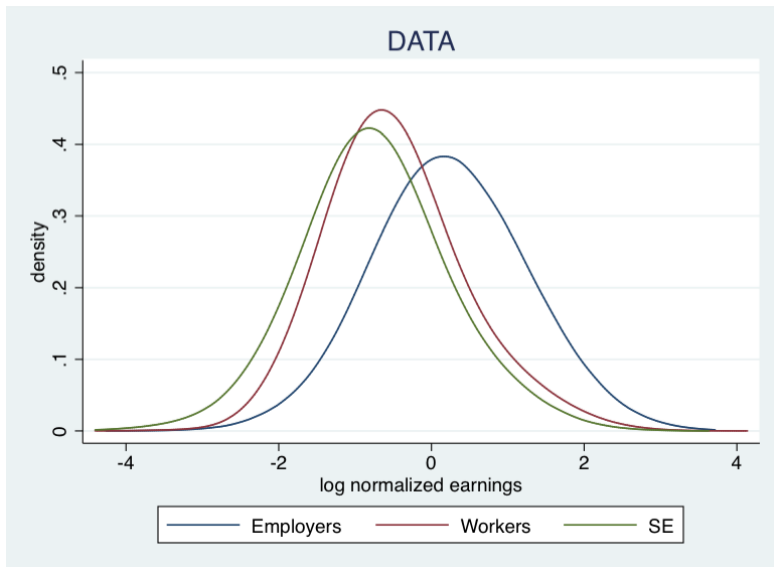
Income Distribution I

- Median entrepreneurial and wage income is not different ...



Income Distribution II

- ...but within entrepreneurs there are two very different groups



What we do:

- Parsimonious extension of the Lucas (1978) model with:
 - ▶ Financial frictions
 - ▶ Heterogeneity in two skills → working and entrepreneurial
 - ▶ Time allocation decision
- Calibrate two versions of the model:
 - ▶ Standard model without worker heterogeneity and time allocation
 - ▶ Full model with worker heterogeneity and time allocation
- Show consequences of financial frictions on:
 - ▶ Occupation structure
 - ▶ Aggregate output and TFP
 - ▶ Inequality

Mechanism

- Without financial frictions, occupational choices **only** depend on comparative advantages

Mechanism

- Without financial frictions, occupational choices **only** depend on comparative advantages
- Financial frictions **distort** these decisions → **Assets** also matter now
 - ▶ Good but poor managers have to start small → self-employed
 - ▶ Rich and mediocre managers become employers
 - ★ Get high return on their capital
 - ★ Benefit from low wages
 - ▶ Also affect the average skill of workers
- Overall, the economy has **more** self-employed and **less** workers

Mechanism

- Without financial frictions, occupational choices **only** depend on comparative advantages
- Financial frictions **distort** these decisions → **Assets** also matter now
 - ▶ Good but poor managers have to start small → self-employed
 - ▶ Rich and mediocre managers become employers
 - ★ Get high return on their capital
 - ★ Benefit from low wages
 - ▶ Also affect the average skill of workers
- Overall, the economy has **more** self-employed and **less** workers
- Two skill heterogeneity allow us to have poor and rich entrepreneurs with and without financial frictions

Preview of findings

1. Two skill heterogeneity is important:
 - ▶ incorporate self-employment into Lucas (1978)
 - ▶ evidence on earnings across occupations
 - ▶ cross-country differences in entrepreneurship are mostly due to self-employment
2. Self-employment: crucial to understand impact of financial frictions
3. Benchmark versus Standard model
 - ▶ Larger impact of financial frictions on occupational structure, output, TFP and inequality
4. Financial frictions affect inequality through many channels:
 - ▶ labor income accounts for 20% of the income variance in the baseline economy (but more than 60% with no financial frictions)
 - ▶ **persistence** of income and **correlation** between capital and labor income increases

The model

Economic Environment

- Closed economy of infinitely-lived households
- Households are heterogeneous in two skills (z_w, z_m) (can be correlated) and in wealth (a)
- Households are endowed with one unit of time
 - ▶ They allocate their time to manage or work $t_m + t_w = 1$
- Production function uses three inputs: Managerial input, Labor and Capital
- Financial intermediation industry is competitive
- Entrepreneurs face a collateral constraint: $k \leq \lambda * a$

Household's Problem: Worker

$$\max_{c_j, a_{j+1}} E \left\{ \sum_{j=1}^{\infty} \beta^j U(c_j) \right\}$$

subject to

$$c_j + a_{j+1} = wz_w + (1+r)a_j$$

$$c_j, a_{j+1} \geq 0,$$

Household's Problem: Entrepreneur

$$\max_{c_j, a_{j+1}} E \left\{ \sum_{j=1}^{\infty} \beta^j U(c_j) \right\}$$

subject to

$$c_j + a_{j+1} = y_j^e$$

$$c_j, a_{j+1} \geq 0,$$

where

$$y_j^e \equiv \max_{n, t_m, k} \{ m^\gamma k^\nu n^\theta - w n^d - r(k - a) + a_j - \delta k - c_f \mathbf{I}_{n^d > 0} \}$$

subject to

$$m = t_m z_m,$$

$$n = (1 - t_m) * z_w + n^d,$$

$$\text{where } t_m \in [0, 1], n^d \geq 0, k \leq \lambda * a$$

Calibration

Calibration Strategy

- ① We calibrate two version of our model economy:
 - Standard model: without time decision and with homogenous worker ability
 - Benchmark model: heterogeneity in both skills and time allocation decision
- ② Set exogenously for both models
 - Utility function $U(c) = \ln(c)$.
 - Parameters of production function $\gamma = 0.198$, $\eta = 0.3256$ and $\theta = 0.4764$ (Guner et. al. 2008).
 - Depreciation rate (δ) is set at 6%.
- ③ We minimize a loss function between moments in the model and in the data
 - Standard: 4 parameters and 4 moments.
 - Benchmark: over identified, 13 parameters and 15 moments

Calibration Strategy: Shock Process

1 Standard model: follow Buera and Shin (2011)

$$- z_{m,i,t} = \rho z_{m,i,t-1} + \sigma_{m,i,t}^2$$

2 Benchmark Model:

- $\ln(z_{w,i,t}) = \alpha_{w,i} + u_{w,i,t}$
- $\ln(z_{m,i,t}) = \alpha_{m,i} + u_{m,i,t} + \zeta_{i,t}$
- $u_{j,i,t} = \rho_j u_{j,i,t-1} + \epsilon_{j,i,t}$ for $j = w, m$
- $\epsilon_t = (\epsilon_{w,t}, \epsilon_{m,t})$ jointly drawn from a bivariate normal with corr. $\rho_{\epsilon_{w,m}}$
- $\zeta_{i,t}$ is drawn from a normal distribution with mean 0 and variance σ_{ζ}^2
- Each period there is a probability that a households draw a new skill from the invariant distribution
- Managerial fixed effect (α_m): drawn from a Standard Pareto distribution with tail parameter η_m
- Working fixed effect (α_w): drawn from a Normal distribution with variance $\sigma_{w,f}^2$ and mean that depends on the realization of α_m

Calibration Strategy: Standard Model

- 1 Serial correlation of managerial shocks to match the persistence of entrepreneurship between two consecutive years
- 2 Variance of innovations to managerial ability to match the variance of log entrepreneurial income
- 3 Discount factor and tightness of financial frictions to match the capital-to-output and credit-to-gdp

Calibration Strategy: Benchmark Model

- 1 Mean fixed effect on managerial ability (μ_m) and fixed cost of employers (c_e) to match the occupational structure
- 2 Var. of working ability shock ($\sigma_{\epsilon_w}^2$) and Pareto tail of fixed eff. on managerial ability (η_m) to match the var. of log-earnings of W and E
- 3 Variance and autocorrelation of managerial skills ($\sigma_{\epsilon_w}^2, \rho_m$), correlation between managerial and working skill innovation ($\rho_{\epsilon_m, w}$) and transitory shock to managerial ability (σ_{ζ}^2) to match the yearly transitions rates across occupations
- 4 Correlation of fixed effects on working and managerial skills ($\rho_{\alpha_m, w}$) and probability of the lowest fixed effect on managerial skill (p_m) to match the difference in mean-log earnings between occupations
- 5 Prob. of drawing new fixed eff. (p_{α}) to match the var. of log consumption
- 6 Discount factor (β) and tightness of financial frictions (λ) to match the capital-to-output and credit-to-gdp

Calibration Results

Panel A. Baseline Model Economy		
p_m	Prob. first fixed effect on manag. skill	0.229
η	Pareto tail managerial skill	5.40
μ_m	Location managerial skill distrib.	-2.976
$\rho_{\alpha_{m,w}}$	Correlation working and managerial fixed effects	0.145
$\sigma_{\epsilon_{w,w}}^2$	Variance innovation working skill	0.073
$\sigma_{\epsilon_m}^2$	Variance innovation managerial skill	1.145
ρ_m	Autocorrelation managerial skill	0.788
$\rho_{\epsilon_{m,w}}$	Correlation innovations managerial and working skills	0.335
σ_{ζ}^2	Variance transitory shock managerial skill	0.212
β	Discount factor	0.886
λ	Collateral constraint	1.27
c_e	Fixed cost of employers	0.08
p_{α}	Probability of new skills draw	0.041
Panel B. Standard Model Economy		
σ_m^2	Variance innovation managerial skill	1.25
ρ_m	Autocorrelation managerial skill	0.929
β	Discount factor	0.898
λ	Collateral constraint	1.29

Calibration Results

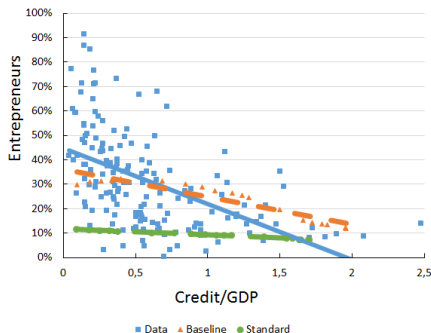
Panel A: Baseline Model Economy		
	Data	Model
K / Y	2.11	2.04
Credit / Y	0.42	0.43
Var(y_w)	0.76	0.76
Var(y_{ent})	1.06	1.07
Avg y_{EMP} -Avg y_W	0.66	0.66
Avg y_{SE} -Avg y_W	-0.29	-0.28
Avg y_{EMP} -Avg y_{SE}	0.94	0.93
Workers	68%	68%
Self-Employed	24%	24%
Employers	8%	8%
W to W	94%	94%
W to SE	5%	6%
SE to SE	77%	75%
SE to E	8%	8%
E to SE	22%	23%
E to E	68%	76%
Var(log(c))	0.80	0.73
Panel B; Standard Model Economy		
K / Y	2.11	2.0
Credit / Y	0.42	0.42
Persistence Entrep.	0.87	0.88
Var(y_{ent})	1.06	1.06

Calibration Results: Non-targeted Moments

	Data	Baseline
<i>Income Transitions</i>		
W to W	0.01	0.01
W to SE	-0.06	-0.34
W to E	0.57	0.83
SE to W	0.00	0.10
SE to SE	-0.03	-0.02
SE to E	0.45	0.74
E to W	-0.37	-0.35
E to SE	-0.65	-0.98
E to E	0.16	0.08
<i>Income Inequality within Entrepreneurs</i>		
$\text{Var}(y_{SE})$	0.869	0.859
$\text{Var}(y_{EMP})$	0.934	1.058
<i>Inequality Measures - Labor Income</i>		
Gini	0.53	0.56
Earnings ratio $p90/p10$	10.7	8.9
Earnings ratio $p75/p25$	3.2	3.5
Between group inequality	4.0%	4.3%
Within group inequality	96.0%	95.7%

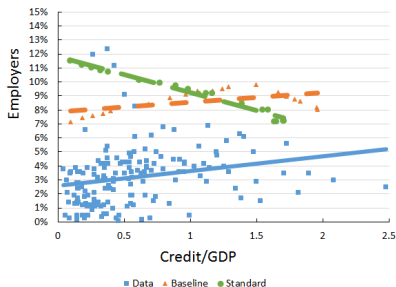
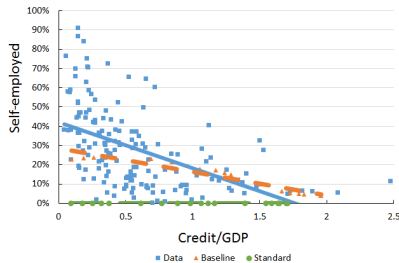
Financial Frictions, Entrepreneurship and the Cross-Country Data

Financial Frictions and Entrepreneurship



- Regression coefficients are 0.23 in the data, 0.11 in the baseline model and 0.025 in the standard model
- The baseline model explains half the variation in the data while the standard model explains one tenth

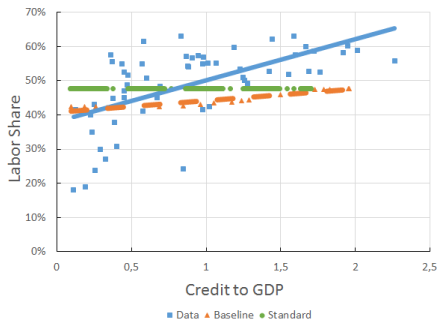
Financial Frictions and Entrepreneurship



- The baseline model can capture the strong negative association between *Credit* and *Self-Employment*

Employee Compensation

- In the baseline model the shift from being self-employed to being employee increase the share of employee compensation in national income
- In the standard model this is given by the Cobb-Douglas parameter and does not change with financial development
- The model can capture 30% of the relation between employee compensation and credit observed in the data



Output and TFP

- We regress Output and TFP on Credit-GDP for the data, baseline model and standard model
- The regression coefficient in the data is 0.34 for output and 0.26 for TFP
- The regression coefficient in the baseline model is 0.23 for output and 0.12 for TFP
- The regression coefficient in the standard model is 0.24 for output and 0.12 for TFP
- The baseline model do not seem to deliver different results than the standard model to explain the differences in cross-country data

Aggregate Effects of Removing Financial Frictions

Aggregate Effects of Removing Financial Frictions

- When we remove financial frictions output increase 58% in the baseline model and 47% in the standard model
- This is explained by the larger increase in *Credit-to-GDP* (1.53) deliver in the baseline economy compare to the standard model (1.28)
- Key for the results is the *larger decrease* in entrepreneurs in the baseline model
 - ▶ Average skill of entrepreneurs increase 200% compare to 50% in the standard model
- TFP increase is also larger in the baseline model (26% vs 21%)

Removing Financial Frictions: Aggregate Effects

	(a) Baseline	(b) Standard
<i>Change in fraction of</i>		
Workers	0.20	0.035
Entrepreneurs	-0.20	-0.035
Self-Employed	-0.20	0
Employers	0.0	-0.035
Credit to GDP	1.53	1.28
<i>% Change</i>		
Output	58%	47%
Avg skill workers	-8%	0%
Avg manag skill entrep.	201%	52%
TFP_{data}	23%	19%
TFP_{ideal}	26%	21%

$$TFP_{ideal} = \frac{Y}{K^{\eta} L^{\theta}}$$

$$TFP_{data} = \frac{Y}{K^{\frac{\eta}{\eta+\theta}}}$$

Removing Financial Frictions: Sensitivity

- We perform 3 robustness:
 - ▶ Homogenous workers with time allocation
 - ▶ Set fixed cost paid by employers to zero
 - ▶ Fixed costs in terms of labor
- Time allocation is important to get the negative relation between entrepreneurs and credit which amplifies the results on output
- Changes in fixed cost do not seem to matter much

	Baseline	Standard	$\sigma_w^2 = 0$	$c_e = 0$	$c_e * w$
<i>Change in fraction of</i>					
Workers	0.20	0.035	0.07	0.22	0.20
Entrepreneurs	-0.20	-0.035	-0.07	-0.22	-0.20
Self-Employed	-0.20	0	-0.09	-0.12	-0.19
Employers	0.0	-0.035	0.03	-0.10	-0.01
Credit to GDP	1.53	1.28	1.29	1.52	1.53
<i>% Change</i>					
Output	58%	47%	54%	55%	57%
Avg skill workers	-8%	0%	0%	-14%	-8%
Avg manag skill entrep.	201%	52%	180%	217%	193%
TFP_{data}	23%	19%	26%	21%	22%
TFP_{ideal}	26%	21%	31%	24%	26%

Financial Frictions and Inequality

Removing financial frictions:

- Reduces the Gini index of assets from 0.91 to 0.84
- Reduces the Gini index of consumption from 0.60 to 0.56
- Reduces the autocorrelation of earnings from 0.89 to 0.80
- Reduces the correlation of assets and earnings from 0.71 to 0.38
- Important effects on sources of inequality

Table: Variance Decomposition of Income

	Baseline Model		Standard Model	
	Baseline	PCM	Baseline	PCM
Capital income	31%	12%	31%	17%
Skill returns	20%	64%	20%	57%
Covariance term	49%	24%	48%	25%

- All this results are present in the standard model but to a lesser extent

Summary

We extend Lucas (1978) to include household heterogeneity in managerial and working skills:

- ① theory accounts for self-employment (or SE)
- ② theory consistent with Brazilian household data on earnings between and within occupation, occupational transitions, and the cross-country variation in occupational structure
- ③ modeling the occupational structure is important for understanding aggregate and distributive effects of financial frictions

**Allub, Gomes y Kuehn(WP, 2020):
Human capital and financial development:
firm-level interactions and macroeconomic
implications**

Economic growth: human capital and financial development

- Two mechanisms largely explore independently to understand growth are:
 - ▶ Accumulation of physical capital (in particular the role of credit)
 - ▶ Accumulation of human capital

Economic growth: human capital and financial development

- Two mechanisms largely explore independently to understand growth are:
 - ▶ Accumulation of physical capital (in particular the role of credit)
 - ▶ Accumulation of human capital
- **This paper:** Studies the interaction of human capital and financial frictions in the presence of capital-skilled complementarities

Economic growth: human capital and financial development

- Two mechanisms largely explore independently to understand growth are:
 - ▶ Accumulation of physical capital (in particular the role of credit)
 - ▶ Accumulation of human capital
- **This paper:** Studies the interaction of human capital and financial frictions in the presence of capital-skilled complementarities
- Capital-skilled complementarity implies a non-trivial interaction between human capital and financial frictions.
 - ▶ Credit constrained firms hire a lower proportion of skilled labour than what's optimal;
 - ▶ High skilled wages reduce firms' desired capital intensity and hence relieve the effective financial constraint.

Objective

Quantify the importance of financial frictions, human capital, and the combination of both, for explaining countries differences in output and wage gap.

- Occupational choice model where individuals with different education endowments and wealth, facing financial constraints, decide to set-up a firm or to work;
- Production uses capital, skilled and unskilled labour as inputs;
- Individuals accumulate assets and can only borrow up to a fraction of those assets;

Objective

Quantify the importance of financial frictions, human capital, and the combination of both, for explaining countries differences in output and wage gap.

- Occupational choice model where individuals with different education endowments and wealth, facing financial constraints, decide to set-up a firm or to work;
- Production uses capital, skilled and unskilled labour as inputs;
- Individuals accumulate assets and can only borrow up to a fraction of those assets;
- Calibrate baseline economy to the US.
- Vary educational attainment and credit to GDP across countries.

Model

Occupational choice Model

- Agents differ in their skill level:
 - ▶ a fraction θ are skilled (s)
 - ▶ a fraction $1 - \theta$ are unskilled (u)
- Agents are endowed with managerial ability $z \in [0, \bar{z}]$ with cdf and densities:
 - ▶ $F(z)$ and $f(z)$
- Each period there is a probability ζ of drawing a new managerial ability
- Agents decide how much to consume and save, which gives an endogenous distribution of assets
- According to their z , a and skill, agents either become employees or entrepreneurs.

Production

- Entrepreneurs profits are given by:

$$\max_{\{n_i^u, n_i^s, k_i\}} \pi(z_i, a_i) = y_i - w^u n_i^u - w^s n_i^s - r k_i.$$

$$\begin{aligned} \text{st } y_i(n_i^u, n_i^s, k_i) &= A z_i^{(1-\gamma)} [\mu (n_i^u)^\sigma + (1-\mu) [\lambda (q k_i)^\rho + (1-\lambda) (n_i^s)^\rho]^{\frac{\sigma}{\rho}}]^{\frac{\gamma}{\sigma}} \\ k &\leq \chi a \end{aligned}$$

- From the FOC wrt K we get $\rightarrow f'_k = r_i = r + \omega_i$
- Working with the skilled-unskilled labor ratio we get:

$$\text{sign} \left(\frac{\partial \left(\frac{n_i^s}{n_i^u} \right)}{\partial r_i} \right) = \text{sign} \left(\frac{\rho - \sigma}{(1-\rho)(1-\sigma)} \right) \quad (1)$$

- Capital skill complementarity implies $\rho < \sigma \Rightarrow \text{sign} \left(\frac{\partial \left(\frac{n_i^s}{n_i^u} \right)}{\partial r_i} \right) < 0$.

The Individuals's problem

Chooses $\{c_i, a'_i, z^{*,e}(a_i)\}$ in order to maximize

$$\sum_{t=0}^{\infty} \beta^t \frac{(c_t^i)^{1-\psi}}{1-\psi}, \quad (2)$$

$$\begin{aligned} c_i + a'_i &= I_{z_i(a_i) < z^{*,e}(a_i)} (w^e + Ra_i) + \\ &+ I_{z_i(a_i) \geq z^{*,e}(a_i)} (\pi(z_i, a_i) + R * \max(0, (a_i - k_i))). \end{aligned}$$

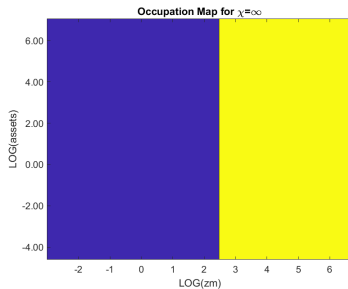
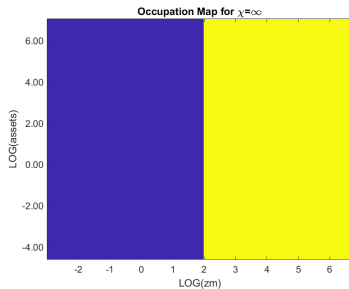
Value functions

For an individual of skill e , ability z_i and endowed with assets a_i , the value of being in occupation O is given by

$$\begin{aligned} V_e^O(z_i, a_i) &= \max_{\{a'_i, c_i\}} \left(U(c_i) + \right. \\ &\quad \left. \beta \zeta E[I_{z_i(a'_i) < z^{*,e}(a'_i)} V_e^{wk}(z_i, a'_i) + I_{z_i(a'_i) \geq z^{*,e}(a'_i)} V_e^{ent}(z_i, a'_i)] + \right. \\ &\quad \left. \beta(1 - \zeta) E[I_{z'_i(a'_i) < z^{*,e}(a'_i)} V_e^{wk}(z'_i, a'_i) + I_{z'_i(a'_i) \geq z^{*,e}(a'_i)} V_e^{ent}(z'_i, a'_i)] \right) \end{aligned}$$

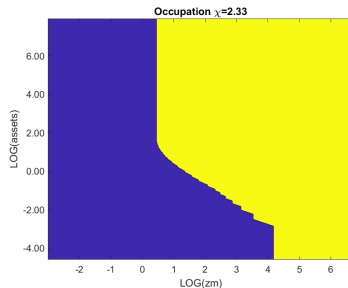
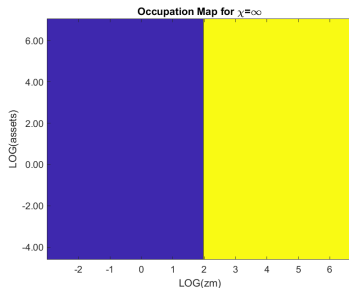
Occupational maps-Effect of Education

- More education shift the ability threshold to become entrepreneur to the right



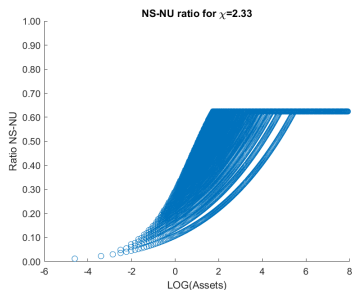
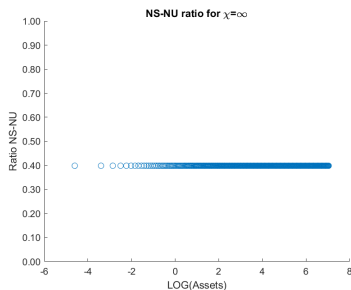
Occupational maps-Effect of Credit

- Financial frictions affect the entrepreneurial decision, allowing rich and mediocre agents to become entrepreneurs, while poor and skillful have to be workers



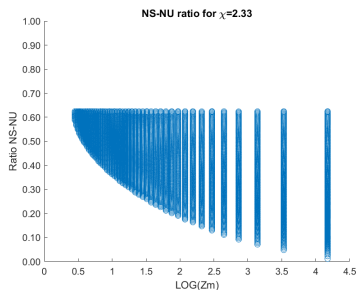
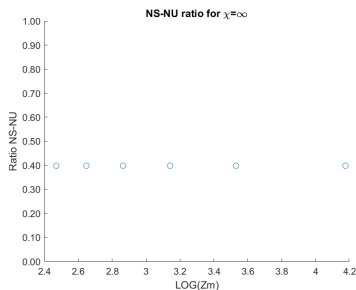
Financial frictions and skill-composition

- More heterogeneity in skill compositions for poor entrepreneurs



Financial frictions and skill-composition

- More heterogeneity in skill compositions for more able entrepreneurs

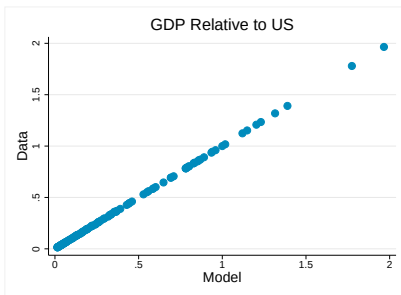
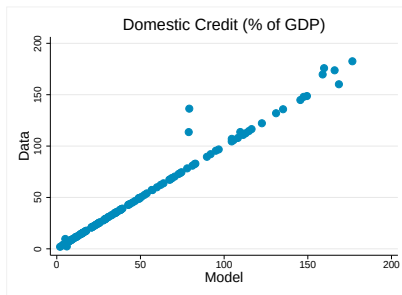


Counterfactual exercises

Quantifying the interactions

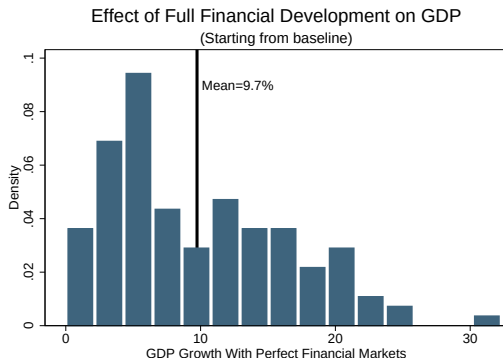
- Economy with different levels of financial frictions, educational attainment (and TFP).
- How much of the GDP differences to the baseline economy ($\theta = 0.28$, $\chi = \infty$ and $TFP = 1$) can be accounted by:
 - ▶ Financial frictions
 - ▶ Education
 - ▶ Interaction

Cross country calibration of χ and TFP



Financial development

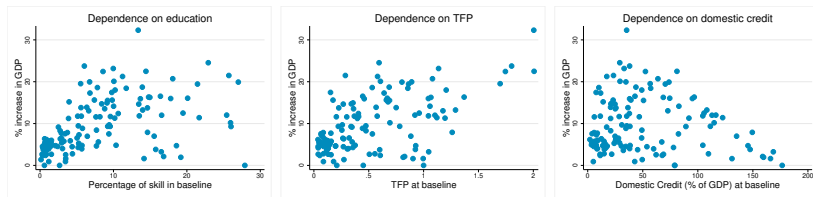
Gains from financial development $\rightarrow \chi = \infty$



Notes: Model calculation. For each country, we calibrate χ_i and TFP_i to match Domestic Credit to GDP and GDP per capita PPP (from World Bank Development Indicators, average over 2000-2009), and we set θ_i equal to the share of college educated in the population above 25 in 2005 from Barro and Lee (2013). We then compute a counterfactual model economy with $\xi = \infty$, and calculate the percentage increase in GDP.

Gains from financial development $\rightarrow \chi = \infty$

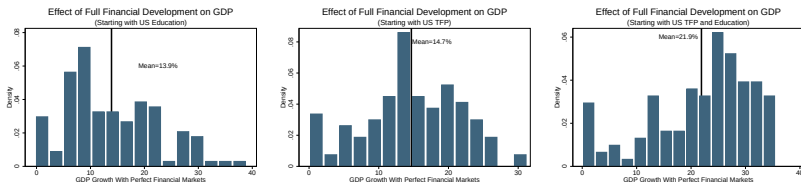
Figure: What determines the gains of financial development?



Notes: For each country, we plot GDP gains of perfect financial markets against the level of education, TFP, and initial domestic credit to GDP. The estimated multivariate regression results in $g_j = 6.01(17.70) + 0.62(16.42)\theta_j + 11.9(20.90)A_j - 0.15(-23.72)Credit_j$, with t-statistics in parenthesis (and R-squared of 0.886).

Gains from financial development $\rightarrow \chi = \infty$

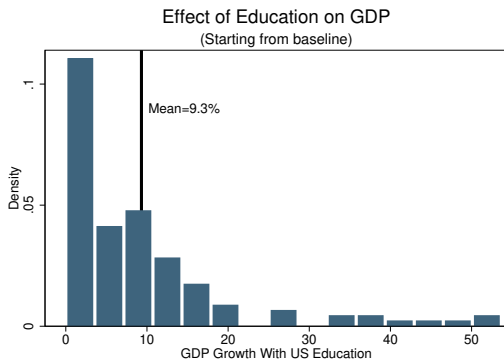
Figure: What determines the gains of financial development?



Notes: Model calculation. For each country, we create counterfactual economies with: (i) θ_{US} , (ii) TFP_{US} and (iii) θ_{US} and TFP_{US} , and compute the counterfactual GDP gains of perfect financial markets in each of the economies.

Human capital development

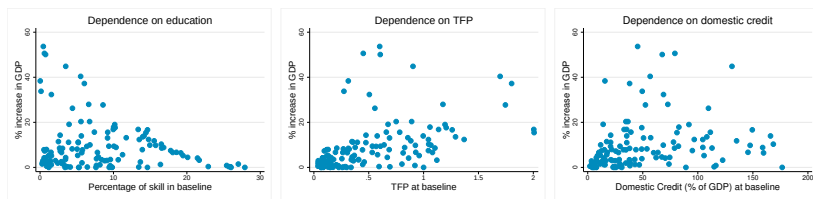
Gains from human capital development $\rightarrow \theta = 0.28$



Notes: Model calculation. For each country, we calibrate ξ_j and TFP_j to match Domestic Credit to GDP and GDP per capita PPP (from World Bank Development Indicators, average over 2000-2009) and we set θ_j , to the share of individuals with college education in 2005 from Barro and Lee (2013). We then compute a counterfactual economy with $\tilde{\theta}_j = \theta_{US}$, and calculate the percentage increase in GDP.

Gains from human capital development $\rightarrow \theta = 0.28$

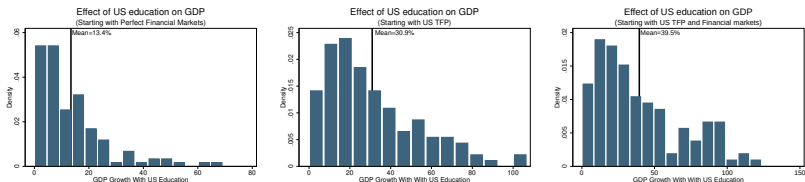
Figure: What determines the gains from educational improvement?



Notes: For each country, we plot GDP gains of perfect financial market again the initial level of education, TFP and domestic credit to GDP. The estimated multivariate regression give $g_j = 6.27(5.61) - 1.22(-9.85)\theta_j + 17.25(9.13)A_j + 0.08(3.87)Credit_j$, with the *t*-statistics in parenthesis (and a *R*-squared of 0.562).

Gains from human capital development $\rightarrow \theta = 0.28$

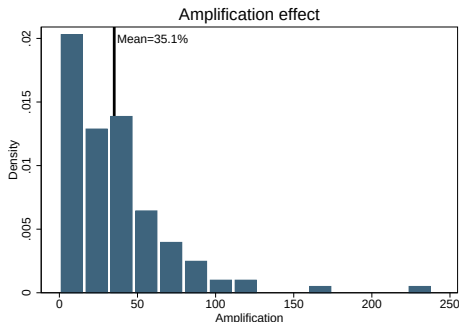
Figure: What determines the gains from educational improvement?



Notes: Model calculation. For each country, we create counterfactual economies with: (i) ξ_{US} , (ii) TFP_{US} and (iii) ξ_{US} and TFP_{US} , and compute the counterfactual GDP gains of raising educational attainment to the US level in each of the economies.

Gains from both reforms together

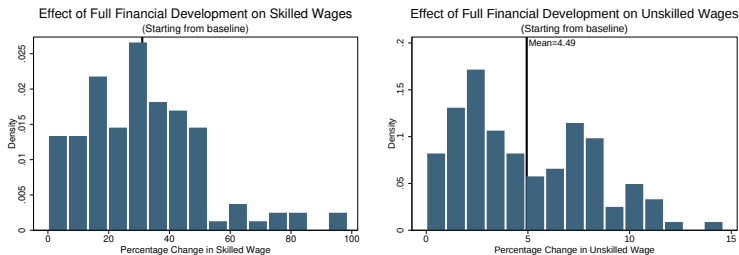
Figure: Amplification effect of simultaneously liberalizing financial markets and increasing education to the US level, all countries



Notes: Model calculation. The amplification effect is calculated as the growth rate of output of implementing $\chi_j = \chi_{US}$ and $\theta_j = \theta_{US}$, divided by the sum of the effects of implementing each one separately minus one.

Effects of reforms - College premium

Figure: Effects of removing financial frictions on wage inequality



Notes: Based on model simulations. Both graphs consider a situation where financial markets are perfect, i.e., $\xi_j = \infty$ for each country. The left-hand graph displays the effect on skilled wages and the right-hand graph on unskilled wages.

Cross-country regressions: model and data

Cross-country firm level

- Regression of percentage of educated workers force by firm from [World Bank Enterprise Survey](#).
- Size of firm measured by the log assets.
- Interaction term with domestic credit to GDP.

Regression of fraction of skilled on size

Table: Firm-level regressions: model and data

	Data	Model
Log of firm's assets	3.872*** [2.54, 5.20]	4.540*** [4.11, 4.97]
Log of firm's assets $\times$ domestic credit/GDP	-0.020** [-0.038,- 0.002]	-0.028*** [-0.034,-0.022]
Country dummies	X	X
Industry dummies	X	
Observations	10,359	5,825,835
R-squared	0.379	0.865

Notes: T-statistics are shown in brackets. *** indicates significance at the 1% level, ** indicates significance at the 5% level, and * indicates significance at the 10% level. In both regressions standard errors are clustered at the country level. Dependent variable is the % of workers with >10 years education. Using simulated data, each firm is weighted by its mass.

- Firms with more assets have larger skilled-unskilled labor ratios
- These relation is weaker as we increase the credit/gdp ratios

Conclusions

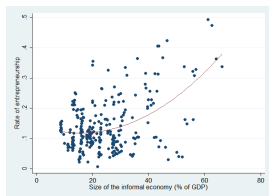
- Capital-skill complementarity and financial frictions imply that constrained firms use different level of skill intensities.
- Gains from an education reform depend on the level of financial development.
- Gains from a financial reform depend on the population level of education.
- There are large amplification effects of performing both reforms together.
- Part of a wider question of “why do financial liberalizations” fail.

Lucila Berniell (BEJMacro):
Occupational Choice and Investments in Human
Capital in Informal Economies

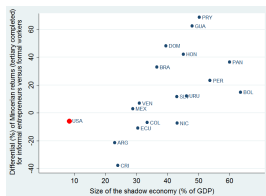
Motivation

Cross country evidence: As the share of the **informal economy** becomes larger

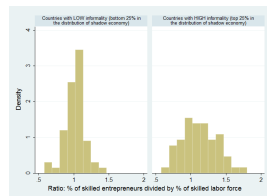
- 1 The rate of entrepreneurship increases [Fact 1].
- 2 The difference between the skill premium for entrepreneurs and for workers becomes larger [Fact 2].
- 3 The fraction of high-skilled individuals that choose to be entrepreneurs is larger [Fact 3].



(a) Fact 1



(b) Fact 2



(c) Fact 3

Figure: Three facts connecting informality, occupations and human capital investments

What I do

- I develop an occupational choice model with human capital investments and an informal sector.
- I produce a number of analytical results showing that the model economy is able to reproduce the three motivating facts.

The model economy

- OLG of two-periods lived agents, with heterogeneous innate ability ($z \in [0, 1] \sim \Gamma(z)$).
- Three factors of production: l_u , l_s , and k .
- Each individual has one unit of time that can be offered in the market for unskilled workers when young and when old.
- Young agents can choose to invest time in the formation of human capital ($t_e \in \{t_e^U, t_e^S\}$, with $0 = t_e^U < t_e^S$) in order to produce $\tilde{z}$ according to

$$\tilde{z} = z^{\theta(\frac{1}{\theta} - t_e)}, \text{ where } 0 < \theta < \frac{1}{t_e^S}. \quad (3)$$

- In the second period:
 - ▶ skilled workers $l_s = \tilde{z}$, and skilled entrepreneurs $\tilde{z} = m$ (managerial skills).
 - ▶ unskilled workers $l_u = 1$, and unskilled entrepreneurs $z = m$.
- Entrepreneurs produce according to $y = m^\gamma F(k, l_u, l_s)^{1-\gamma}$ (with capital-skill complementarity and $F(k, l_u, l_s)$ constant returns to scale).
- Preferences: $U(c_1, c_2) = \log c_1 + \beta \log c_2$.

The model economy: Informal and formal managers, and financial intermediation

- Managers can be formal (taxed at $\tau > 0$) or informal ($\tau = 0$).
- Managers can:
 - ▶ self finance part of the physical capital with savings from first period,
 - ▶ borrow from financial intermediaries (at rate r , exogenous), and can choose to default paying $\eta > 0$ ($\eta = 0$ for informal).
- Imperfect contract enforcement: $0 < \eta < 1$.
- Gross profits of entrepreneurs:

$$\Pi(k, m) = \max_{l_u, l_s \geq 0} m^\gamma F(k, l_u, l_s)^{1-\gamma} - (1+r)k - w_s l_s - w_u l_u.$$

- Net profits:

$$\begin{aligned} W(a, m; \eta, \tau) &= \max_{s \leq a, d \geq 0} (1-\tau)\Pi(s+d, m) \\ \text{s.t. } (1-\tau)\Pi(s+d, m) + a(1+r) &\geq (1-\eta)(1-\tau)[\Pi(s+d, m) + (s+d)(1+r)] \\ &\quad + (a-s)(1+r). \end{aligned} \tag{4}$$

- **Remark:** Informal managers are **excluded** from financial intermediation (they cannot fulfill the ICC).

The model economy: Decisions

- Individual with ability z solves

$$V(z) = \max_{a \geq 0, t_e \in \{t_e^U, t_e^S\}, \phi_w \in \{0,1\}, \phi_f \in \{0,1\}} \log(c_1) + \beta \log(c_2), \quad (5)$$

subject to

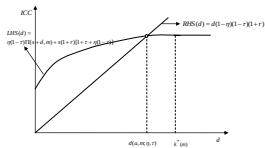
$$c_1 = (1 - t_e)w_u - ct_e - a,$$

$$c_2 = a(1 + r) + \phi_w \left[\left(1 - \frac{t_e}{t_e^S}\right) w_u + \frac{t_e}{t_e^S} w_s \tilde{z}(z, t_e) \right] + \\ (1 - \phi_w) [\phi_f W(a, \tilde{z}(z, t_e); \eta, \tau) + (1 - \phi_f) W(a, \tilde{z}(z, t_e); 0, 0)],$$

$$\text{and } \tilde{z} = z^{\theta(\frac{1}{\theta} - t_e)}.$$

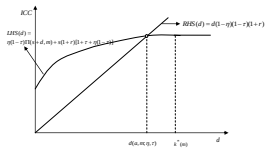
Characterization of the equilibrium: The demand for credit

Characterization of the equilibrium: The demand for credit

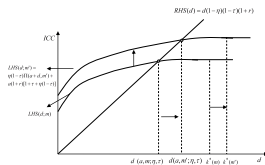


(a) ICC and credit demand.

Characterization of the equilibrium: The demand for credit

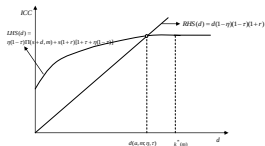


(a) ICC and credit demand.

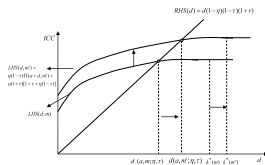


(b) Loan increasing in m .

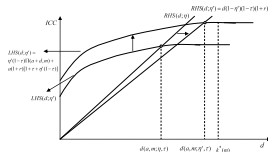
Characterization of the equilibrium: The demand for credit



(a) ICC and credit demand.



(b) Loan increasing in m .



(c) Loan increasing in η .

Characterization of the equilibrium: Occupational and educational choices

Four possible configurations of cutoffs, satisfying:

- UW at the bottom and UFM at the top of the ability distribution.
- Formal are abler than informal.
- Skilled informal managers are abler than unskilled informal managers.

Characterization of the equilibrium: Occupational and educational choices

Four possible configurations of cutoffs, satisfying:

- UW at the bottom and UFM at the top of the ability distribution.
- Formal are abler than informal.
- Skilled informal managers are abler than unskilled informal managers.

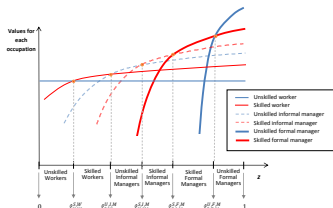


Figure: Occupational, educational and sector choices (Case 1)

Results: Higher costs of contract enforcement

Results: Higher costs of contract enforcement

- More costly contract enforcement $\Rightarrow$ more informal managers ($\downarrow \eta \Rightarrow \uparrow$ informality in the economy).
- When η decreases (informality rises):
 - ▶ The use of capital by informal managers is not directly affected (they were already excluded from credit).
 - ▶ The value of formal managers is reduced (they are more credit constrained).
 - ▶ Capital-skill complementarity $\Rightarrow \downarrow$ demand for skilled labor ($\downarrow$ value skilled workers).

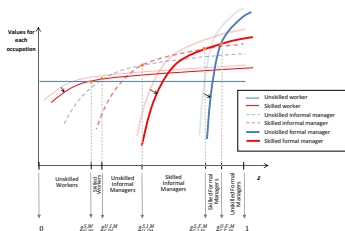


Figure: Higher costs of enforcing contracts (lower η): Changes in the cutoffs.

Results: Higher costs of contract enforcement

With a lower η (higher informality):

- There are more entrepreneurs (Fact 1).
- The value for skilled workers is reduced while the value for informal skilled managers is not (Fact 2).
- Investments in skills are more concentrated among entrepreneurs (Fact 3).

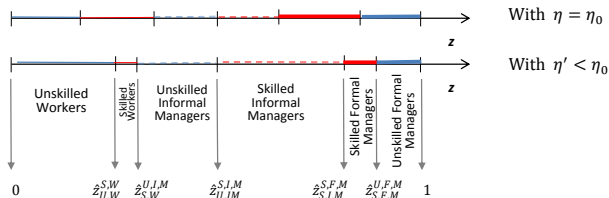


Figure: Occupational cutoffs for two distinct values of η (η_0 and η' , with $\eta_0 > \eta'$).

Conclusions

- The extent of the informal economy can affect the incentives to invest in human capital.
- Evidence on three facts linking informality, occupational choices and educational attainment.
- Analytical framework to explain these facts.
- Mechanism: Under capital-skill complementarity, more informality (due to higher costs of contract enforcement) affects occupational and educational choices since the investment in skills becomes relatively less valuable for workers than for entrepreneurs.